and, subsequently, the most advanced techniques for investigating short-lived radionuclides are described.

17.1. GENERAL CONSIDERATIONS

Radioactive nuclides may be prepared by a wide variety of techniques e.g. particle accelerators and nuclear reactors; however, only cyclotrons or reactors of at least moderate particle flux are able to produce sources of sufficiently high specific radioactivity to be of practical interest. These two methods of production complement each other since, in general, they do not produce the same isotopes of an element.

In cyclotrons charged particles such as protons, deuterons, and α-particles bombard the target nuclei, and after emission of one or more particles to remove the excess excitation energy, a radioactive product nuclide may result. In the capture of positively charged particles and the subsequent emission of neutrons the product radionuclides are *neutron deficient* compared to the stable isotopes of the element; see Fig. 4.1 and 5.7. Another important point in cyclotron bombardments is that normally the product is not isotopic with the target. As a result, after chemical separations a product of high specific activity is obtained since it is not diluted by the target material. An example of an important cyclotron-produced radionuclide is ^{22}Na ($t_{½}$ 2.6 y) formed by the reaction:

$$^{24}_{12}\mathrm{Mg} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{22}_{11}\mathrm{Na} \tag{17.1}$$

This contrasts with the production of radionuclides by neutron irradiation, e.g. in a nuclear reactor, where the neutron absorption and subsequent de-excitation by emission of γ-rays produces *neutron rich* isotopes, frequently radioactive, which are isotopic with the target. A typical reaction is

$$^{23}\mathrm{Na}\ (\mathrm{n}, \gamma)\ ^{24}\mathrm{Na} \tag{17.2}$$

yielding another important isotope, ^{24}Na ($t_{½}$ 15.0 h), which is diluted by the ^{23}Na target atoms. Even if the same radioactive nuclide can be made simpler in a nuclear reactor, cyclotron production may be favored if high specific activity is necessary for a particular use.

A second mode of production in reactors is by the fission process itself as the majority of fission products are radioactive and cover a wide range of atomic numbers of varying abundances. These fission products may be isolated with high specific activity in contrast to the neutron capture products. However, for the isolation of any particular radioactive element from the rest of the fission products can require a fairly lengthy separation; we discuss this further in Ch. 21.

Radioactive nuclides may also be obtained as daughters of parent nuclides which have been previously made either by cyclotron or reactor irradiation or exist naturally (e.g. the production of ^{234}Th from ^{238}U). Several examples of such radioactive isotope generators are discussed in §5.16.

Which mode or production is utilized depends on several factors, e.g. the availability of accelerators, reactors etc., the availability of starting material, the length of time for production, separation and purification processes, the time from production to shipping, and the shipping time determine the half-life with which it is practical to work. The type of experiment and equipment available for handling and measurement determines the preferable decay type and intensity is also important. All these factors must be considered in determining which nuclide is to be used. For example, if in an experiment requiring sodium tracer these aspects indicate the necessity of a time of at least several weeks, the use of reactor-produced ^{24}Na is excluded due to its short half life and cyclotron-produced ^{22}Na is required. If the experiment requires the use of very high specific activities, either cyclotron or fission product activities may be necessary. Reactor activities are usually much cheaper and easier to obtain on a routine basis because many targets can be irradiated simultaneously in the reactor whereas in a cyclotron only one target at a time can be bombarded.

Although these considerations are important in the choice of a radionuclide to use for a particular purpose, the possibilities of production may also be limited by the reaction cross-sections. Figure 5.7 (right side) shows the reaction paths needed to transform a given target nuclide into a product nuclide. If the target nuclide is denoted by ${}^{A}_{Z}\mathrm{X}$, a (n,γ) reaction yields the same product as a (d,p) reaction, i.e. ${}^{A+1}_{Z}\mathrm{X}$. The choice of reaction depends on the particular nuclides available, projectile energy, and the cross-sections (available in charts or tables of nuclear data). As pointed out in Chapter 11, the same projectile may yield several different products depending on e.g. its energy; e.g. Fig. 11.2 shows six different α-induced reactions.

An induced nuclear reaction may be followed by rapid radioactive decay, leading to the desired radionuclide. Thus a (n,2n) reaction on ${}^{A}_{Z}\mathrm{X}$ may yield ${}^{A-1}_{Z}\mathrm{X}$, which rapidly decays to ${}^{A-1}_{Z+1}\mathrm{X}$ through β^--emission. For example, the formation reaction

$$^{238}\mathrm{U}\,(\mathrm{n},2\mathrm{n})\,^{237}\mathrm{U} \tag{17.3}$$

followed by

$$^{237}\mathrm{U} \xrightarrow[6.75\ \mathrm{d}]{\beta^-} {}^{237}\mathrm{Np} \tag{17.4}$$

These reactions occur in nuclear rectors and can used to produce ^{237}Np for investigation of neptunium chemical behaviour. An alternative method for obtaining ^{237}Np is to use ^{241}Am. The americium is deposited on a column and its daughter nuclide, ^{237}Np, is "milked out", by a suitable leachant.

17.2. IRRADIATION YIELDS

The production of a radioactive nuclide by irradiation in a reactor or an accelerator can be written as:

$$X_{target} + X_{proj.} \xrightarrow{k} X_1 \xrightarrow{\lambda_1} X_2 \xrightarrow{\lambda_2} X_3 \tag{17.5}$$

We are mainly concerned with the product X_1 (produced at rate k), which is assumed to be radioactive, decaying in a single step or via a chain of decays to a final stable nuclide. The number of radioactive product atoms N_1 of X_1 present at any time t is the difference between the total number of product atoms formed and the number of radioactive decays that have occurred in time dt. This leads to the differential equation

$$dN_1 = k\,dt - \lambda_1 N_1\,dt \tag{17.6}$$

Integration between the limits of $N_1 = 0$ (this assumes that no radioactive product nuclide X_1 exist at the beginning of the irradiation) and N_1 corresponding to an irradiation time t_{irr}:

$$N_1 = (k/\lambda_1)\left(1 - e^{-\lambda_1 t_{irr}}\right) \tag{17.7}$$

N_1 increases with time towards a maximum value, which is reached when $t_{irr}\lambda_1 \gg 1$ (i.e. $t_{irr} \gg t_{½}$):

$$N_1(\max) = k/\lambda_1 = k\,t_{½}/0.693 \tag{17.8}$$

However, for $t_{irr} \gg t_{½}$, by expanding the exponential into a MacLaurin series, $e^{-\lambda_1 t_{irr}} \approx 1 - \lambda_1\, t_{irr}$, which with (17.7) gives

$$N_1 = k\,t_{irr} \left(\text{for } t_{irr} \ll t_{½}\right) \tag{17.9}$$

If we measure the irradiation time in number of half-lives, $a = t_{irr}/t_{½}$, we obtain

$$N_1 = N_{1,max}\left(1 - 2^{-a}\right) \tag{17.10}$$

This equation shows that irradiation for one half-life ($a = 1$) produces 50% of the maximum amount, two half-lives give 75%, etc.

After the end of the irradiation time, the radioactive product nuclides continue to decay according to their half-lives. We indicate this decay time after the end of irradiation (termed the cooling time) as t_{cool}. The number of product nuclides, which is a function of the irradiation time t_{irr} and cooling time t_{cool}, is given by

$$N_1 = (k/\lambda_1)\left(1 - e^{-0.693 t_{irr}/t_{½}}\right) e^{-0.693 t_{cool}/t_{½}} \tag{17.11}$$

The effect of the cooling time is shown in Figure 17.1 for two sample nuclides.